

CATEGORICAL OVERVIEW OF WHAT I KNOW OF ALGEBRAIC GEOMETRY

ARTHUR ZANETTI¹

1. VARIETIES

Although, at a first look, there are four different types of varieties, there really is only one. Well first and foremost affine varieties are quasi-affine varieties and projective varieties are quasi-projective varieties. Second of all we can easily embed $\mathbb{A}^n$ into $\mathbb{P}^n$ via the map

$$\begin{aligned}\phi : \mathbb{A}^n &\longrightarrow \mathbb{P}^n \\ (x_1, \dots, x_n) &\longmapsto (1 : x_1 : \dots : x_n)\end{aligned}$$

Say now $X \subset \mathbb{A}^n$ is an affine-variety and $X' \subset X$ is open in X . Then $I(\phi(X))$ is actually the homogenization of the members of $I(X)$ and so $\overline{\phi(X)} = Z(I(\phi(X)))$ is a projective variety (I didn't check the details, but it feels right). Then $\phi(X')$ is open in $\phi(X)$ since ϕ is an embedding and $\phi(X)$ is open in $\overline{\phi(X)}$ hence $\phi(X')$ is open in $\overline{\phi(X)}$. So every quasi-affine variety is a quasi-projective variety and we may reduce all of our analysis, at least when it comes to this broad overview, to quasi-projective varieties. I will say **variety** from now on to refer to quasi-projective varieties.

2. THE VARIETY CATEGORY

Even though we already used morphisms above when saying ϕ was an embedding, I wanna take the time to go through a definition that will work in any variety. Let Y be a variety over an algebraically closed field k , we say that a function $f : Y \rightarrow k$ is **regular** at $P \in Y$ if $\exists U \subset Y$ open with $P \in U$ such that $f|_U = h/g$ for polynomials $h, g \in k[x_1, \dots, x_n]$, $\exists n \in \mathbb{N}$ and with $Z(g) \cap U = \emptyset$. If Y is quasi-projective we require in addition that h and g be homogeneous of same degree. We say that f is regular if it is regular at every point of Y . If X is another variety we will say a function $\psi : X \rightarrow Y$ is a **morphism** if it is continuous under the Zariski topologies and $\forall V \subset Y$ open and $q \in O(V)$ (the ring of regular functions on V) we have $(q \circ \psi) \in O(\psi^{-1}(V))$.

From these definitions we form the first category of large interest in algebraic geometry, that of varieties whose morphisms are what we defined above. I will call this category **Var** from now on. As far as I understand classical algebraic geometry can be summarized as the study of properties invariant under isomorphism in **Var**.

From my knowledge there is no category equivalence between **Var** and some other usual category. However we have an important result with a subcategory of **Var**

Theorem 2.1. *The association $X \rightarrow A(X)$ between affine varieties and domains that are finitely generated as k -algebras is an arrow-reversing category equivalence.*

3. THE BIRATIONAL CATEGORY

There is a category in which we use the same objects but has a looser requirement for morphisms. We start with an example. When one blows-up $\mathbb{A}^n$ on 0 we obtain a variety $Bl_0(\mathbb{A}^n) \subset \mathbb{A}^n \times \mathbb{P}^{n-1}$ that isn't isomorphic to $\mathbb{A}^n$. However if we denote by $E \subset Bl_0(\mathbb{A}^n)$ the exceptional subset then $Bl_0(\mathbb{A}^n) \setminus E$ is an open dense subset (all non-empty open sets are dense under the Zariski topology) and $Bl_0(\mathbb{A}^n) \setminus E$ is actually isomorphic to $\mathbb{A}^n \setminus 0$. This leads us to define a notion of equivalence of varieties that requires that varieties only be isomorphic on open dense subsets.

Let Y be a variety, we will say $f : Y \dashrightarrow X$ is **rational map** to mean that f is a set of pairs $\langle U, g \rangle$ where $g : U \rightarrow X$ is a morphism and the morphisms agree on the intersection of their domains. We will say that the domain of f is the union of all domains of the morphisms in the pairs $\langle U, g \rangle$. Note that in general we cannot compose rational maps so if we want to create a category using these kinds of maps we will need to add further restrictions. The problem is precisely that if $h : X \dashrightarrow Z$ is another rational map then it is possible that f maps its domain into a set disjoint from the domain of h . We will say that a morphism $g : U \rightarrow X$ is **dominant** if $\overline{g(U)} = X$. We will say f is dominant if any (hence all) morphisms that define f are dominant. Now we can compose f and h to get a rational map $Y \dashrightarrow Z$. Notice how we don't even need h to be dominant for the composition to make sense. We will denote by **Bir** the category of varieties where the morphisms are dominant rational maps. We say two varieties are **birationally equivalent** when they are isomorphic in **Bir**.

One might rightfully argue that this notion of equivalence doesn't mean much, that removing a set with empty interior from the variety changes it in a fundamental way. From everything I know of algebraic geometry up till 09/09/2025 I'm inclined to agree. I will give another example to illustrate just how fundamental of a change removing a set with empty interior can be. It is trivial that the map $\mathbb{A}^n \rightarrow \mathbb{P}^n \setminus Z(x_n)$ such that $(x_1, \dots, x_n) \mapsto (1 : x_1 : \dots : x_n)$ is an isomorphism. Since $Z(x_n) \subset \mathbb{P}^n$ has empty interior, under our new definition $\mathbb{P}^n$ and $\mathbb{A}^n$ would then be equivalent. However $O(\mathbb{A}^n) = k[x_1, \dots, x_n]$ and $O(\mathbb{P}^n) = k$ so their rings of regular functions couldn't be more different! The ring of regular functions carries most of the information I know how to extract from a variety so this notion of equivalence feels too loose.

Nonetheless there are other notions which are invariant under birational equivalence. The strongest result I'm aware of in this direction is

Theorem 3.1. *The association $X \mapsto K(X)$ that maps varieties to their function fields is an arrow-reversing equivalence of categories between **Bir** and the category of finitely generated field extensions of k .*

So most likely I just lack knowledge of what there is of interesting to study under this new equivalence.

REFERENCES